WASP-50b

R-1045

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_161204` · created 2026-07-30T09:03:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457726.797 +/- 0.034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.051
Transit depth (Rp/Rs)^2	2.92 +/- 1.73 [%]
Orbital Inclination (inc)	87.13 +/- 2.93
Airmass coefficient 1 (a1)	0.43 +/- 0.25
Airmass coefficient 2 (a2)	0.54 +/- 0.34
Scatter in the residuals of the lightcurve fit is	57.89 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.5
Transit Duration (day)	0.078 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.051 vs 0.1404 (deviation 0.6σ)
Duration fit vs published	0.078 d vs 0.0754 d (deviation 0.13σ)
Mid-time O–C	-19.7 min
Depth SNR	3.4
Residual scatter	57.89 %

Light curve

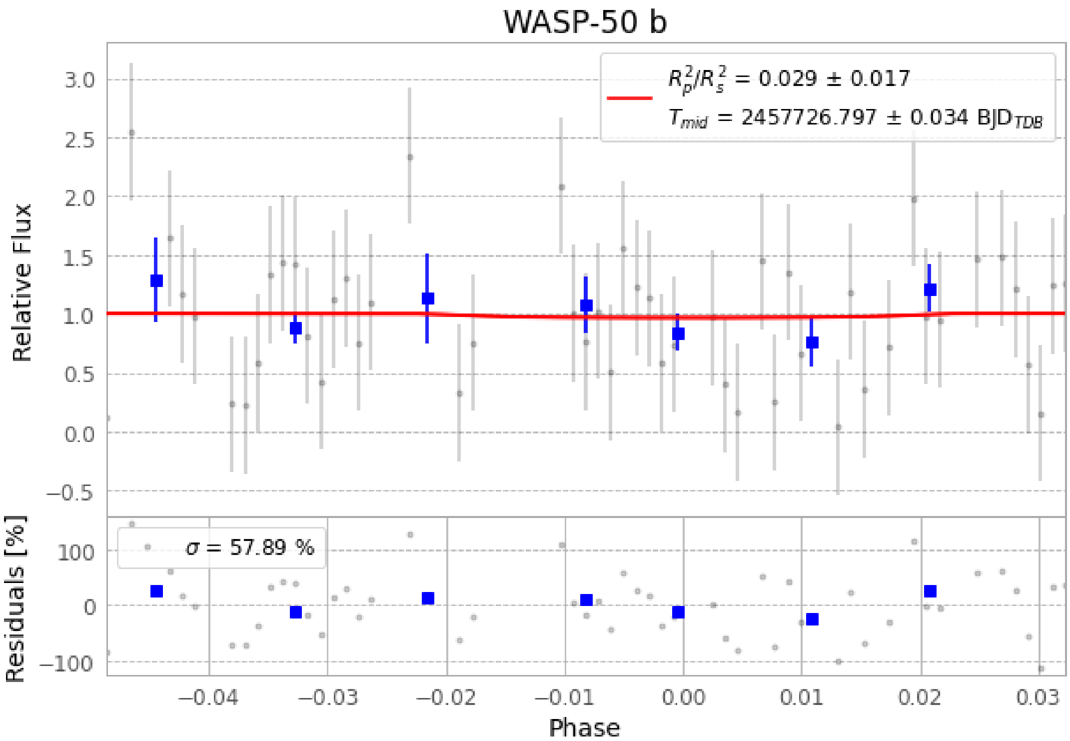

Figure 1 — FinalLightCurve_WASP-50 b_2016-12-03.png (SHA-256 90143eee0e7d0127…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 146], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 da3b037facd26dc6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

78 FITS frames (51 MB), `WASP-50161204044212.FITS` → `WASP-50161204083312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-161204.log (ef145acef0b7…), FinalLightCurve_WASP-50 b_2016-12-03.png (90143eee0e7d…), FinalLightCurve_WASP-50 b_2016-12-03.csv (cebdde5b93b2…)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)